Battery storage optimization on the French power market

Day-ahead arbitrage, price uncertainty, risk management (CVaR, drawdown) and primary reserve (FCR) for a **50 MW / 100 MWh** battery, on French market data **2023-2025**. Every setting is chosen on 2024; 2025 is an out-of-sample test.

2025 (out of sample)	Profit	
Perfect foresight, arbitrage only	€1.76M	theoretical ceiling
Deterministic (ARX forecast)	€1.23M	70% of the ceiling
Risk-averse, mean-CVaR (λ = 0.375)	€1.21M	−2% profit, +18% on the 10% worst days
Arbitrage + FCR, λ = 0.375	€3.64M	including the battery's own impact on the FCR price

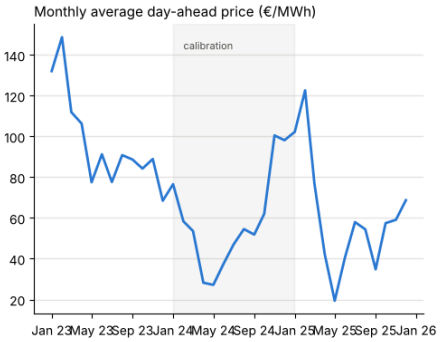
Takeaways. Risk sits on **calm days**, not crisis days. Risk aversion is cheap and statistically significant. FCR dominates revenue, but the battery itself lowers the French FCR price by about a quarter: ignoring it would overstate the 2025 revenue by 25% (€0.9M).

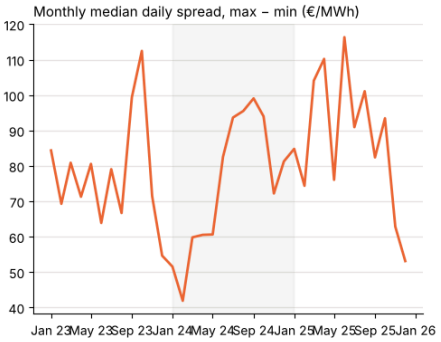
1. Data and protocol

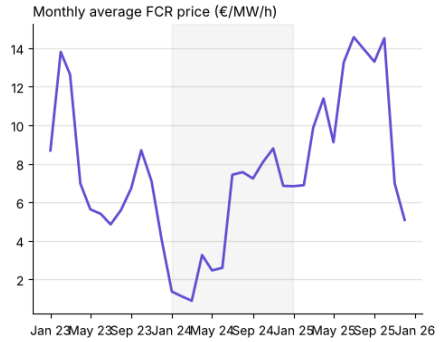
French day-ahead prices from Energy-Charts (CC BY 4.0, source Bundesnetzagentur | SMARD.de; 15-minute prices since October 2025 averaged to hourly). FCR results and anonymous bid lists from regelleistung.net.

Period	Role
2023	warm-up (forecast history)
2024	**calibration**: every setting is chosen here
2025	**test**: settings frozen

year	Mean price (€/MWh)	Negative hours	Median daily spread (€/MWh)	FCR, 1 MW all year (k€)
2023	97	147	77	66
2024	58	352	76	42
2025	61	513	86	92







Prices fell after the crisis but **daily spreads stay high** (€76-86/MWh): a battery earns the spread, not the level. Negative hours multiply with solar (147 → 513). FCR collapsed in 2024 as batteries flooded the market, then partly recovered in 2025.

2. Battery model and perfect-foresight benchmark

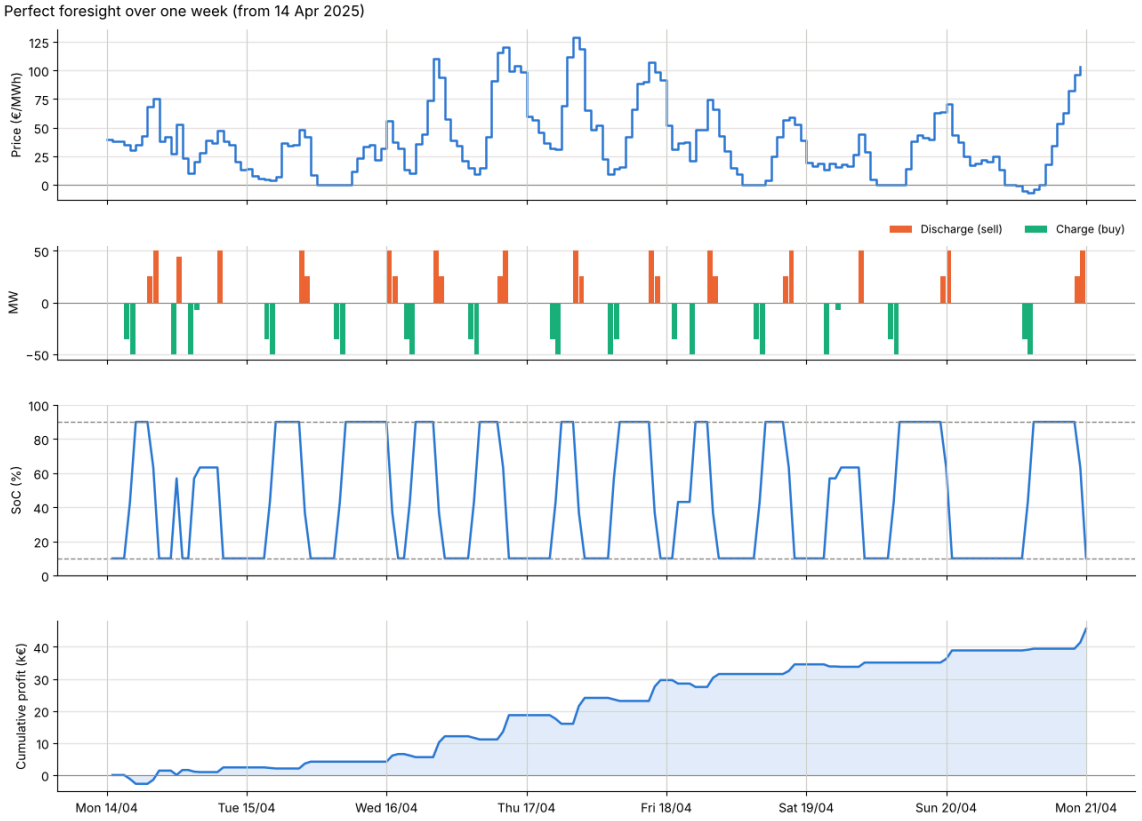
50 MW, 100 MWh, state of charge 10-90% (80 MWh usable), round-trip efficiency 88%, back to 50% at the end of each day. Wear is a **linear cost per MWh discharged**: €100k/MWh replacement / (7,000 cycles × 0.8) ≈ **€17.9/MWh**.

MILP (PuLP + HiGHS), for each hour t: charge c_t, discharge d_t, energy e_t, binary b_t:

$$\max \sum_t p_t(d_t - c_t) - c_{deg} d_t \quad \text{s.t.} \quad e_{t+1} = e_t + \sqrt{\eta}\, c_t - d_t/\sqrt{\eta}, \quad c_t \le P b_t,\ d_t \le P(1-b_t)$$

The binary forbids charging and discharging at once (otherwise profitable with negative prices).

year	Daily horizon (k€)	Yearly horizon (k€)	Gain of the yearly horizon
2024	1,383	1,661	+20%
2025	1,760	2,078	+18%



The ceiling is **€1.38M in 2024 and €1.76M in 2025**. The rest of the project keeps the daily horizon, which matches the day-ahead auction (a yearly horizon would add 18-20%).

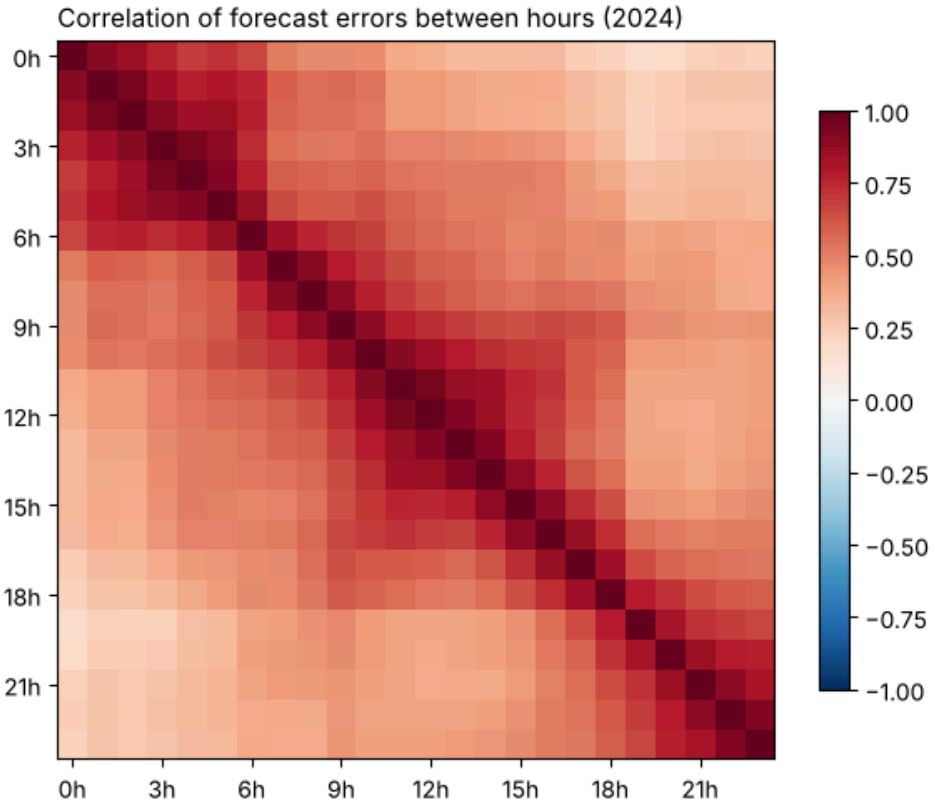
3. Price uncertainty and CVaR optimization

Forecast. On day J before noon, all prices up to the end of J are known, none of J+1. One linear regression per hour, re-estimated daily on 56 days (ARX):

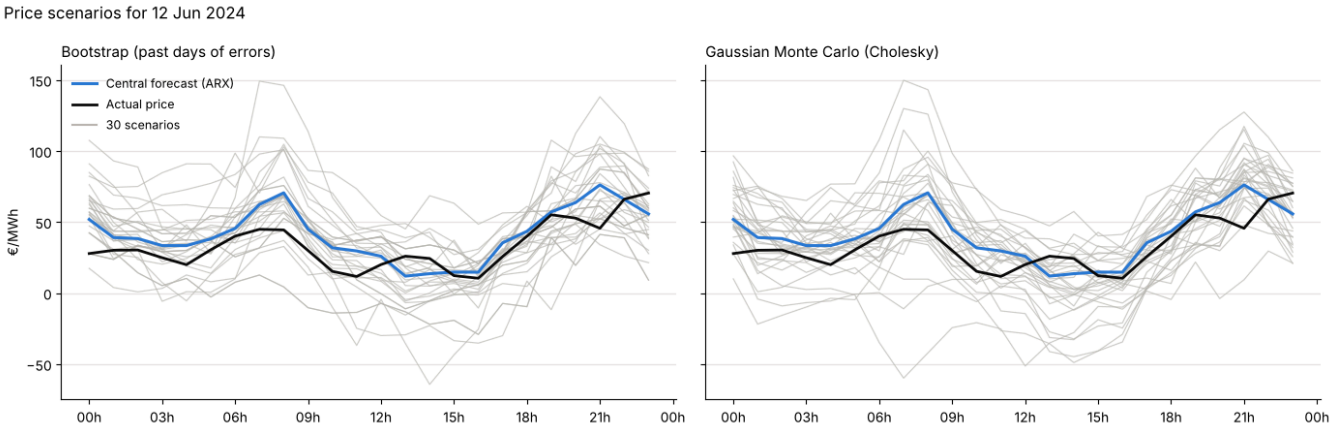
$$\hat{p}_{J+1,h} = \beta_0 + \beta_1 p_{J,h} + \beta_2 p_{J-1,h} + \beta_3 p_{J-6,h} + \beta_4 \min p_J + \beta_5 \max p_J + \text{weekday dummies}$$

It beats the naive forecast ("tomorrow = today") by 14% on 2024 (MAE €17.4 vs €20.4/MWh).

Scenarios. Forecast errors are strongly correlated between hours (2024, figure below): 0.89 on average between neighbouring hours and 0.54 over all pairs, never below 0.17. Two blocks stand out, the night (0.84 within 0-5 h) and the solar hours (0.80 within 10-16 h), linked to each other at about 0.45. A day forecast too high at 10 h is very likely too high at 14 h too: drawing hours independently would give jagged days and understate the risk of a whole bad day. So whole days are simulated: a multivariate normal fitted on the last 60 days of errors, sampled with **Cholesky** ($\Sigma = LL^\top$, $Lz \sim \mathcal{N}(0, \Sigma)$). A bootstrap of past days gave the same quality; the Gaussian version was kept because it can draw any number of distinct days.

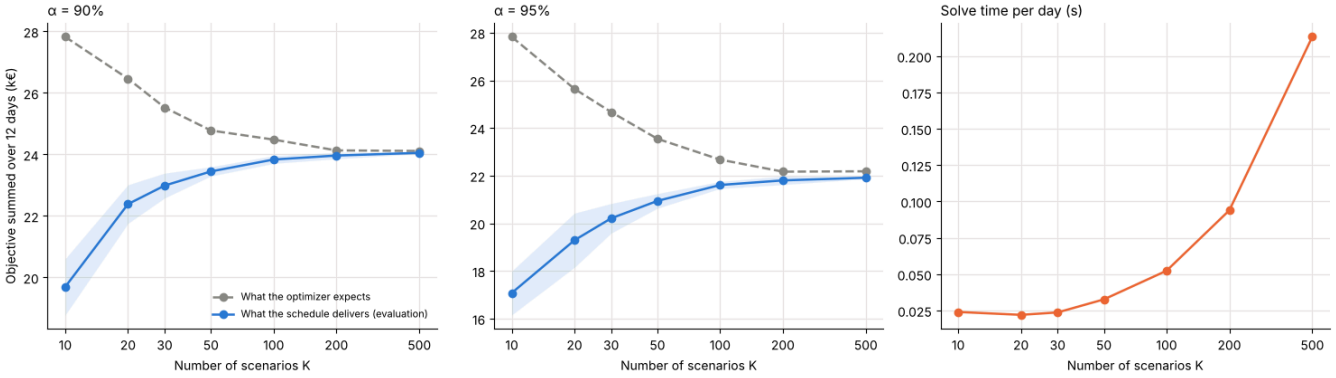


method	Coverage of the 90% interval	Actual spread	Scenario spread
bootstrap	87.0%	76	91
gaussian	88.1%	76	93



Coverage is correct (87-88%), but scenarios **overestimate the daily spread by about 20%**, which makes the optimizer slightly optimistic.

How many scenarios? Schedules optimized on K scenarios, then paid on 5,000 fresh draws: the optimizer's optimism falls from 11% (K = 30) to 0.7% (K = 200). **Kept: K = 200, α = 90%.**



Mean-CVaR. One schedule for all scenarios (decided before prices are known). CVaR 90% = mean of the 10% worst profits, linearized (Rockafellar-Uryasev):

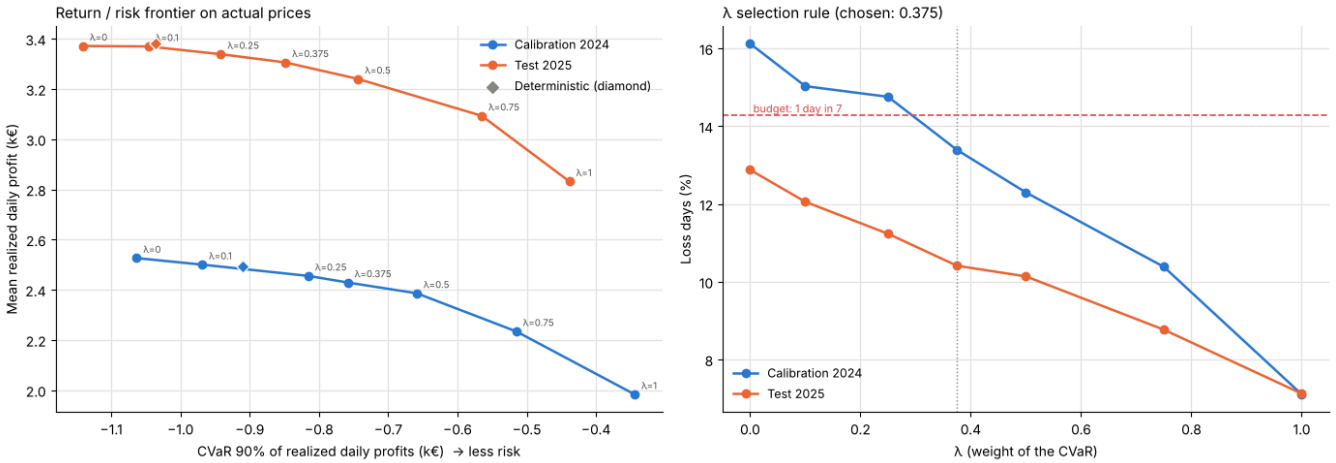
$$\max\ (1-\lambda)\,\mathbb{E}[\pi] + \lambda\Big(\eta - \tfrac{1}{(1-\alpha)K}\sum_s u_s\Big), \qquad u_s \geq \eta - \pi_s,\ u_s \geq 0$$

With λ = 0 it is exactly the optimization on the mean scenario (profit is linear in prices): stochastic optimization only pays once **risk** is priced.

4. Out-of-sample evaluation

Every day of 2024-2025, each strategy decides with the information of the day before and is **paid at actual prices**. **λ rule, fixed in advance:** the most profitable λ on 2024 with at most 1 loss day in 7. Grid: $\lambda \in \{0, 0.1, 0.25, 0.375, 0.5, 0.75, 1\}$.

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Lambda chosen on 2024: 0.375
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In 2024, λ ≤ 0.25 exceed the budget (15-16% of loss days); **λ = 0.375 is the most profitable one that respects it (13.4%)**. The grid was refined with 0.375, the midpoint between 0.25 and 0.5, where the rule switches; the rule itself is unchanged and only 2024 is used to choose. In 2025, λ = 0.375 stays within the budget (10.4%).

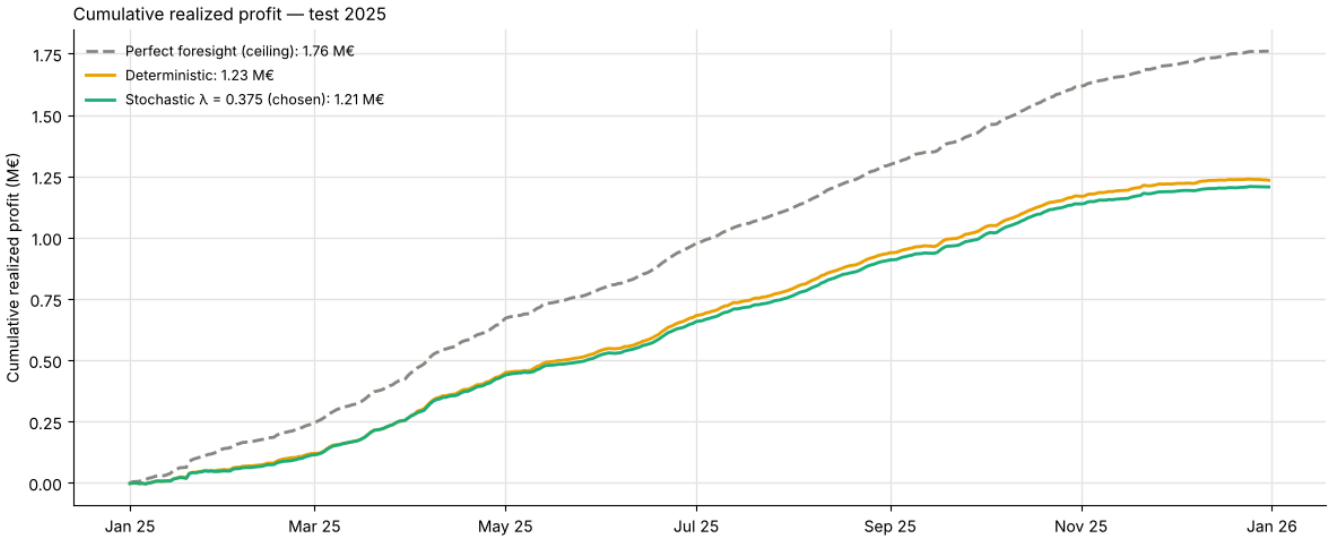
	Profit (k€)	Share of ceiling	Loss days	Daily CVaR 90% (€)	Max drawdown (k€)	Cycles
Perfect foresight	1,760	100%	0%	627	0.0	518
Deterministic	1,234	70%	13%	-1,036	4.7	527
Stochastic λ = 0.375	1,207	69%	10%	-849	4.0	460

4.1 What the stochastic strategy gains over the deterministic one

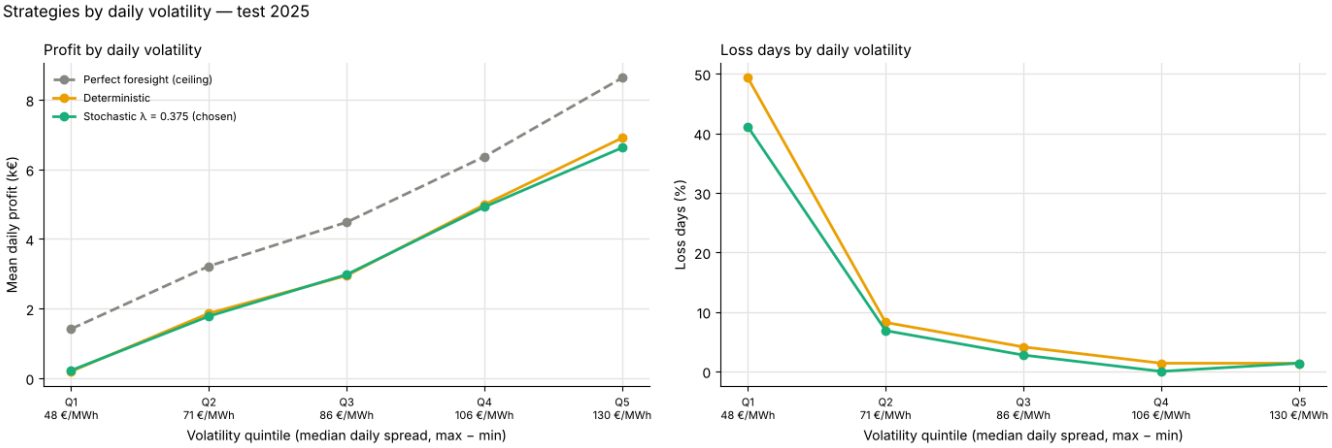
Differences λ = 0.375 minus deterministic, with 95% intervals from a moving-block bootstrap (7-day blocks, same days for both strategies).

	Profit (k€)	Profit (%)	Daily CVaR 90% (€)	CVaR (%)	Loss days (pts)	Cycles (%)
Arbitrage, 2024 (calibration)	-24 [-48; +2]	-2.6%	+152 [+29; +310]	+17%	-2.7	-11%
Arbitrage, 2025 (test)	-27 [-61; +0]	-2.2%	+187 [+9; +316]	+18%	-2.5	-13%
Arbitrage + FCR, 2024	+22 [-5; +51]	+1.3%	+226 [+111; +313]	+93%	-1.4	-19%
Arbitrage + FCR, 2025	+27 [-6; +56]	+0.7%	+170 [+49; +259]	+8%	+0.0	-27%

- **Arbitrage only**: λ = 0.375 gives up about **2% of profit** (−€24k in 2024, −€27k in 2025), a difference that is **not significant** (the interval reaches zero), and improves the **daily CVaR by 17-18%** (+€152 and +€187), which is **significant**. It also has 2.5 points fewer loss days and 11-13% fewer cycles, hence less wear.
- **With FCR**, risk aversion costs nothing (+0.7% to +1.3% profit, not significant) and still improves the CVaR significantly: certain FCR revenue is the hedge the CVaR term looks for.



4.2 Where is the risk?



Losses concentrate on calm days. In the calmest fifth of 2025 (median spread €48/MWh), the deterministic strategy loses money **one day in two** (49%); λ = 0.375 brings this down to 41%. On such days the forecast error is as large as the spread itself. Profit comes from volatile days: the 20% most volatile make about 40% of the annual profit.

4.3 Drawdown budget (side test)

Variant: risk-neutral schedule with a daily floor $\mathrm{CVaR}_{90\%} \geq -(\text{budget} - \text{current drawdown})$, budget = X% of the peak cumulative profit (floor €5k), X ∈ {0.5%, 1%, 2%} chosen on the calibration period (most profitable budget never breached). In 2024-2025 the floor binds on less than 1% of days, so it behaves like λ = 0. Replayed on the **2022 crisis** (same Energy-Charts prices; λ and X re-chosen on May-Dec 2021 with the same rules, test 2022, no FCR):

	Profit 2022 (k€)	Max drawdown (k€)	Longest time below peak (days)	Daily CVaR 90% (€)
Deterministic	1,577	54.6	73	-4,352
Stochastic λ = 0.375 (chosen)	1,725	11.2	14	-2,445
Drawdown budget 2.0%	1,713	13.3	19	-2,870

In 2022 the 2% budget cuts the maximum drawdown from €54.6k to €13.3k and earns 9% more than the deterministic strategy. **But λ = 0.375 alone does as well or better** (€11.2k, 14 days below peak instead of 19, better CVaR), and the choice of X is weak: in 2024 the three budgets give exactly the same result. The variant is therefore not used further.

5. Extension: primary reserve (FCR)

FCR is a symmetric capacity product sold in 4-hour blocks the day before at 8 am; only capacity is paid. For each block, reserving r_b MW requires $c_t + r_b \le P$, $d_t + r_b \le P$ and 15 minutes of full activation in both directions: $E_{min} + 0.25\,r_b \le e_t \le E_{max} - 0.25\,r_b$. FCR revenue is the same in every scenario.

The battery moves the FCR price. The auction is pay-as-clear with a fixed demand, and France is often at its export limit (30% of its demand), so its price is set by French bids alone. Replaying each auction from the anonymous bid lists with the battery offering q MW at €0 gives a French price curve $p_b(q)$ for every block (it gives back the published price in every block of 2023-2025).

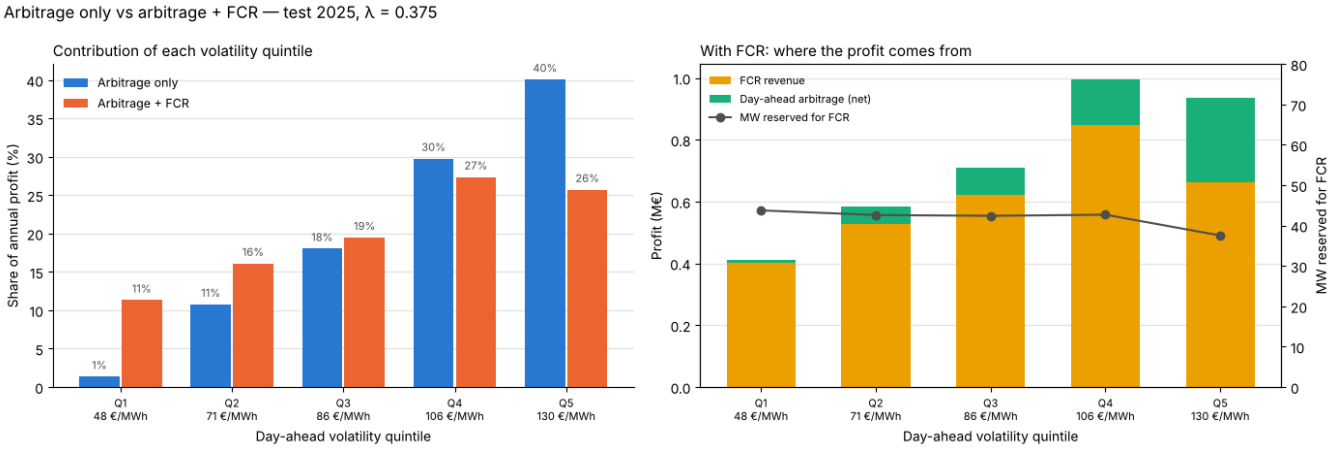
- **Decision**: naive FCR forecast (same block the day before) × factor f of month M−2, with $f = \sum_b p_b(50) / \sum_b p_b(0)$. Bids are only published monthly, so M−2 is the latest month surely available (the 2023 bid lists give the factors of January-February 2024).
- **Evaluation**: reserved MW are paid $p_b(q)$, interpolated from the replayed auction.

day	Blocks where France is at its export limit	FCR only, 50 MW, price taker (k€)	FCR only, 50 MW, with price impact (k€)	Impact
2024	82%	2,116	1,357	-36%
2025	50%	4,600	3,419	-26%

Year	Strategy	Arbitrage only (k€)	Arbitrage + FCR (k€)	Same MW if price taker (k€)	FCR share	Avg MW in FCR
2024	**Perfect foresight**	1,383	2,276	2,687	50%	28
	FCR only	—	1,357	2,116	100%	50
	Stochastic λ = 0.375	889	1,725	2,214	64%	34
2025	**Perfect foresight**	1,760	4,146	4,976	75%	39
	FCR only	—	3,419	4,600	100%	50
	Stochastic λ = 0.375	1,207	3,636	4,556	84%	42

- **The battery's own impact cuts FCR revenue by 36% in 2024 and 26% in 2025** (FCR only, 50 MW: €4.60M → €3.42M in 2025). The impact is strongest when France is at its export limit (82% of blocks in 2024, 50% in 2025).

- **With the impact, stacking clearly beats FCR only** (€3.64M against €3.42M in 2025, +6%; +27% in 2024): the optimizer keeps 34-42 MW in FCR on average and uses the rest for arbitrage. Price-taker figures (right column) would have suggested reserving almost everything for FCR.
- FCR still brings **84% of the 2025 revenue** and loss days almost disappear (0.3-1.4%). With FCR, perfect foresight is only a reference (it decides with the price at full power), not a strict ceiling.



FCR makes revenue much less dependent on volatility: the calmest fifth of days brings 11% of the 2025 profit with FCR, against 1% with arbitrage only.

6. Conclusions and limitations

1. A simple forecast captures **about 70% of the theoretical ceiling**.
2. Stochastic optimization only pays once risk is priced: **λ = 0.375 trades 2% of profit (not significant) for an 18% better CVaR (significant)**, fewer loss days and less wear.
3. Risk sits on **calm days**, where the forecast error is as large as the spread.
4. A **drawdown budget** divides the 2022 drawdown by 4 against the deterministic strategy, but a well-chosen λ does as well without an extra rule.
5. **FCR dominates revenue, but the battery moves the French FCR price** (−26% at 50 MW in 2025). Accounting for it changes the strategy: part of the power is better used for arbitrage.

Limitations. Forecast from past prices only, scenarios too jagged (+20% spread) and too thin-tailed. FCR impact is static (other bidders do not react) and activation energy is neglected. Daily horizon, hourly prices, linear degradation. No intraday or aFRR.

Possible improvements. Forecast with fundamentals (load, wind, solar, gas) and quantiles, plugged into this backtest; aFRR capacity (hourly, asymmetric); intraday as recourse.